

Class 28 Functions

Goals

1. Know the definition of function and inverse function.
2. Know what it means for a function to be one-to-one and onto.
3. Know what it means for two functions to be equal.
4. Be able to determine if a given relation is a function, is one-to-one, is onto.

Definitions and Terminology

Definition: Let D and Y be nonempty sets. A function $f: D \rightarrow Y$ is

Examples: Let $D = \{1, 2, 3, 4\}$ and $Y = \{2, 4, 6, 8\}$. Make up two examples of functions from D to Y . Make up two examples of mappings from D to Y that are **not** functions.

Definition: If f is a function mapping D to Y with range R , then the inverse function for f is a function g mapping R to D that satisfies

Definition: A function f mapping D to Y is one-to-one means

A one-to-one function is also called an injection.

Definition: $f: D \rightarrow Y$ is onto means

An onto function is also called an surjection.

A function that is both one-to-one and onto is called a bijection.

Definition: Functions f and g are equal means

Examples: Decide if each function is one-to-one, onto, both, or neither.

$f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$

$g: \mathbb{R}^+ \rightarrow \mathbb{R}$ defined by $g(x) = \ln(x)$

$f : \mathbb{C} \rightarrow \mathbb{R}^2$ defined by $f(a + bi) = a^2 + b^2$.

$g : M_{2,2} \rightarrow \mathbb{R}$ defined by $g\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$

Practice: prove the following fact about onto functions.

If f and g are onto, then $g \circ f$ is onto.